Auctions with Risk Reserves for Scarce AI Research

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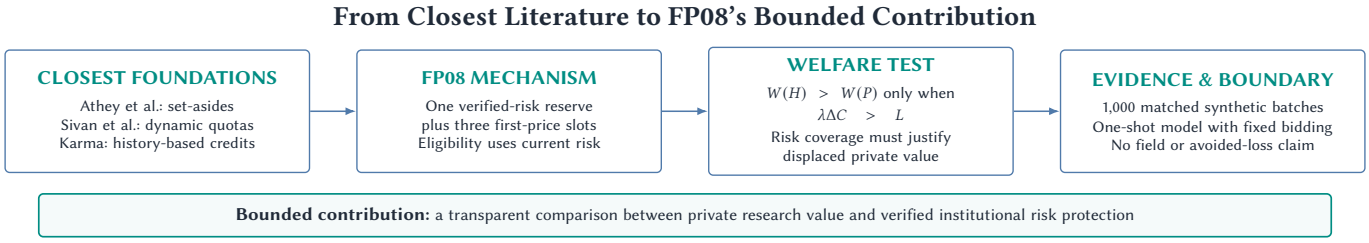


Figure 1: Intellectual-merit comparison showing how FP08 builds on auction set-asides, dynamic quota pricing, and history-based resource allocation. FP08's bounded contribution is a verified-risk reserve that makes the tradeoff between displaced private value and additional risk coverage explicit. The supporting evidence is synthetic and does not establish field effectiveness.

Submission to: COMSCI/ECON 206 — Computational Microeconomics.
Course session: Autumn 2026 Session 1. **Team:** FP08. **Final deadline:** 21 October 2026, 23:00 Asia/Shanghai.
Course instructor: Professor **Luyao Zhang**.

1 Background

Sheng et al. [4] study GenAI adoption and investment performance, not internal access rules. **Stokely** et al. [7] study compute-quota markets; **Sivan** et al. [6] already incorporate heterogeneous values and organizational priorities into dynamic ML-resource pricing. **Pathak** et al. [3] develop non-price reserve systems for medical rationing. Building on these dimensions, we examine a specific unresolved design choice: verified risk reserves alongside paid bidding for indivisible AI research slots, with behavioral sensitivity. This is an application-level comparison, not a claim to invent markets or reserves; medical objectives do not transfer to finance.

2 Motivation

During earnings season, portfolio managers may bid according to private research value even when some requests carry broader verified downside risk for exposed portfolios. PM willingness to pay may neglect firm-level risk priorities, affecting PMs, risk teams, and clients. The decision is whether risk coverage justifies displaced private value.

3 Application Scenario

T. Rowe Price's September 2026 expansion of Claude for portfolio managers (PMs) and analysts motivates our setting [8]. We model an active asset manager during earnings season: 20 PMs compete for four AI research slots. Scarcity and allocation rules are research assumptions, not documented T. Rowe Price practices. In an earnings-season deployment, PMs, risk teams, clients, compliance staff, and computing operations are affected.

4 Research Question

Q1—Economics: When does risk priority justify displaced private value?

Q2—Computation: How do reserve size and value–risk alignment change allocation, payments, utility, efficiency, and risk coverage versus pure first-price auction and FCFS?

Q3—Behavior: With payoffs fixed, does loss framing change bids enough to alter the mechanism comparison?

Shared question: Can auction allocation with a verified risk reserve improve scarce AI research allocation relative to Pure FPA, and under what strategic, computational, and behavioral conditions? Figure 1 links these tasks; behavioral evidence remains planned.

5 Method

Environment and rules. In each batch, $N = 20$ risk-neutral PMs compete for $K = 4$ identical AI research slots. PM i has private value $v_i \in [0, 100]$ and a frozen public risk score $r_i = 100E_iS_i$. Pure FPA allocates four slots by sealed first-price bids; Hybrid reserves one free slot for the highest-risk request with $r_i \geq r^* = 60$ and auctions the remaining capacity; FCFS allocates by exogenous arrival. Utility is $u_i = x_iv_i - p_i$.

Strategic and welfare model. Bayesian private-value bidding provides the strategic benchmark [2, 12], but $b(v) = (n-1)v/n$ applies only to the separate single-slot case; the multi-slot simulation uses stylized bids. For allocation A, firm welfare is

$$W(A) = V(A) + \lambda C(A),$$

where $V(A) = \sum_{i \in A} v_i$ and $C(A) = \sum_{i \in A} r_i$. Thus Hybrid beats Pure FPA if and only if $\lambda \Delta C > L$, where $\Delta C = C(H) - C(P)$ and $L = V(P) - V(H)$.

Computation. We simulate 1,000 matched batches with independent $v_i \sim U[0, 100]$, $E_i, S_i \sim U[0, 1]$, $R = 1$, $\lambda = 1$, and stylized bids $b_i = 0.8v_i$, holding each batch fixed across mechanisms. Full algorithms, sensitivity designs, and reproducibility checks appear in Appendix A.

Artifacts. Code, the runnable notebook, behavioral Space, and poster are linked in the Open Science statement and appendices.

6 Results

Baseline and conditional adoption. Across 1,000 matched synthetic batches, Hybrid loses 26.82 units of private value but gains 36.59 units of risk coverage relative to Pure FPA, yielding mean $\Delta W = +9.78$ at $\lambda = 1$ (MCSE = 0.89). Only 43.8% of batches strictly improve. The break-even weight is $\lambda^* = 0.733$; at $\lambda = 0.5$, mean $\Delta W = -8.52$. Hybrid is therefore conditionally, not universally, preferred.

Robustness. In one fixed batch, increasing marginal PM 11's shading from $0.8v$ to $0.9v$ changes the Hybrid allocation and moves ΔW from -15.85 to $+4.50$, showing sensitivity near allocation margins. Separately, moving PM 20's observed risk score from 59.9 to 60.0 in Batch 713 activates the reserve and changes the winning set, motivating score validation and appeal procedures. These are sensitivity checks, not evidence of a multi-slot equilibrium or realized financial-loss reduction; full outputs appear in Appendix A.

7 Intellectual Merits

Athey, Coey, and Levin examine auction set-asides and subsidies [1]; Sivan et al. use dynamic pricing to allocate scarce machine-learning capacity [5]; and Karma uses credits and allocation history under repeated demand [13]. Our bounded contribution is to compare a pure first-price auction with a hybrid mechanism that reserves one of four AI-research slots for the highest-risk eligible request:

$$W(H) > W(P) \iff \lambda \Delta C > L,$$

where ΔC is the additional verified risk coverage and L is the displaced private value. The evidence comes from 1,000 matched synthetic batches. It does not establish equilibrium bidding, avoided financial loss, or field effectiveness.

8 Practical Impacts

The mechanism affects portfolio managers, risk teams, computing operations, and clients. Relative to the pure auction, the hybrid increases mean synthetic risk coverage from 101.35 to 137.95 and eligible access from 20.29% to 66.70%, while sacrificing private value. These results support intended links to SDG Targets 9.5 and 16.6, but do not demonstrate reduced financial loss [10, 11]. Score gaming, classification error, and biased access remain possible harms. We therefore propose independent score validation, threshold testing, decision logs, appeals, and a shadow deployment. The complete impact pathway appears in Figure 2.

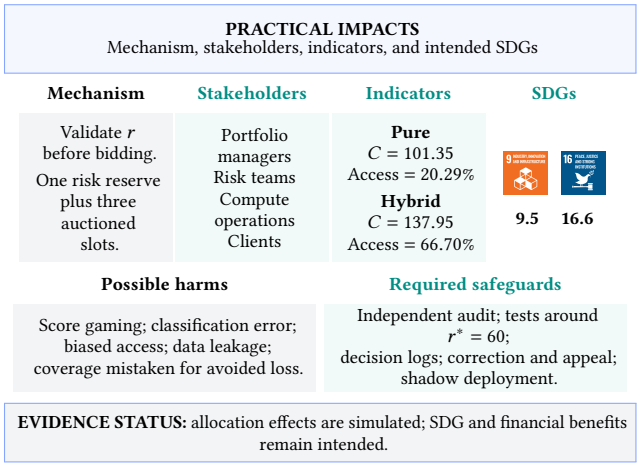


Figure 2: Proposed impact pathway, indicators, risks, safeguards, and intended links to SDG Targets 9.5 and 16.6.

Future Research Roadmap

Figure 3 converts three current limitations into next studies: endogenous multi-slot bidding (economics), risk-score validation and correction (computer science), and a gain/loss framing experiment (behavioral science). Each study specifies its question, method/data, collaborator, progress criterion, and milestone before any deployment claim.

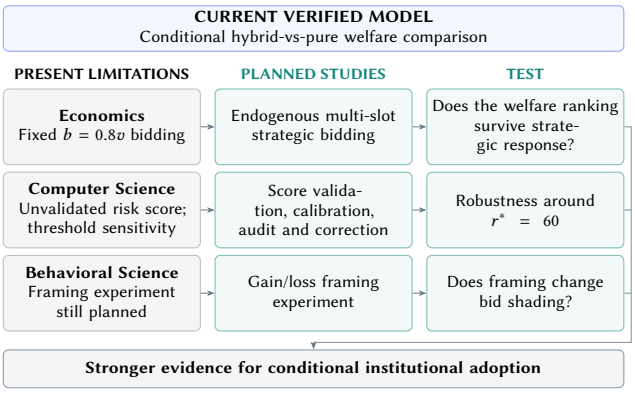


Figure 3: From current limitations to planned tests of conditional adoption.

Evolving 2056 Contribution Statement

We aspire to develop validated theory and auditable infrastructure for allocating scarce AI capacity under private demand and verifiable risk, supported by formal results, replication, field evidence, and correction mechanisms.

Open Science and SDG Statements

The computational repository is available on GitHub, with a runnable Colab notebook and a planned behavioral interface on Hugging Face. Reproduction details appear in Appendix A.2. The project has intended links to SDG Targets 9.5 and 16.6 through research-capacity access and auditable allocation. These are intended, not demonstrated, impacts; full indicators, risks, and safeguards appear in Practical Impacts and Appendix G.

Author Notes

Full Acknowledgments

We thank Prof. **Luyao Zhang** for guiding the development of the project from its strategic research question through its game-theoretic, computational, behavioral, and global-leadership components, and for her formal course feedback. We thank Prof. **Kenneth S. (Ken) Rogerson** and **Han Zhang** for their discussions and feedback that contributed to the development of the project.

We thank **Temur Akhtamjonov** and **Gihun Lee** for their peer reviews, particularly their comments on poster communication, figure presentation, and interpretation of the welfare threshold. We also thank **Longfei Jing** and **Jiayang Sun** for identifying the limitation of the assumed multi-slot bidding rule and for raising questions about endogenous and human bidding behavior. These comments led us to further qualify the computational claims, document sensitivity analysis, and separate completed simulation evidence from planned behavioral evidence.

We thank the COMSCI/ECON 206 classroom community for questions, discussion, and collaborative feedback during the symposium. We also acknowledge Python 3.9.6, Hugging Face Spaces, and OpenAI ChatGPT/Codex as described separately in Appendix A.3. The authors independently reviewed all model assumptions, code outputs, interpretations, and revisions and remain responsible for any remaining errors.

Feedback, classroom participation, or a program role does not automatically imply coauthorship. The authors remain responsible for the proposal, linked artifacts, and remaining errors.

Updated Team and Individual Contributions

Joint intellectual contribution. **Zhenning Wang** and **Qianshuo (Aaron) Wang** jointly developed the research question of when a verified-risk reserve can improve the allocation of scarce AI research capacity relative to a pure first-price auction. The team specified the actors, private values, public risk scores, timing, allocation and payment rules, and the firm objective $W = V + \lambda C$. We jointly narrowed the contribution from a broad claim about AI-quota markets to an application-level comparison of FCFS, pure first-price auction, and a hybrid reserve mechanism. We also agreed to distinguish current risk coverage from avoided financial loss, internal payments from firm welfare, and completed computational evidence from planned behavioral evidence. The resulting proposal connects the analytical condition $\lambda\Delta C > L$, matched simulation results, the Hugging Face decision interface, and the planned gain/loss framing test.

Zhenning Wang. **Zhenning Wang**'s principal intellectual decision was to preserve the boundary between the completed computational comparison and the planned behavioral evidence, and to express that boundary through an inspectable decision interface. The resulting output is the public Hugging Face Space at https://huggingface.co/spaces/dku-comsci-econ206-2026/Final_Zhenning_Qianshuo_PM_Bid, together with PS2 Sections 4–5 and the behavior-and-impact components of the poster. **Zhenning** verified that the Space uses the same $N = 20$, $K = 4$, $R = 1$, $r^* = 60$, assumed bid rule, mechanism sequence, and welfare accounting as the proposal. The Space also labels self-reflection and peer play as exploratory rather than population-level evidence. Following symposium feedback, **Zhenning** recorded the behavioral and practical-impact comments, classified them as Keep, Revise, or Investigate, and checked the Space, paper, and poster for consistency.

Qianshuo (Aaron) Wang. **Qianshuo Wang**'s principal intellectual decision was to formulate the mechanism comparison as the measurable trade-off $\lambda\Delta C > L$ and to separate the single-slot equilibrium benchmark from the assumed multi-slot simulation bids. His inspectable output is the computational repository at https://github.com/dku-comsci-econ206-Autumn2026/PS2_FP8_Qianshuo_Zhenning, including the simulation, runnable notebook, verification script, and archived outputs. Six tests and a clean rerun reproduced the reported output files and primary mechanism-level results. This work connects the strategic model in PS2 Sections 1–2 to the computational answer in PS2 Section 3 and provides the rational benchmark used in the behavioral artifact. Following symposium feedback, **Qianshuo** recorded the behavioral and practical-impact comments, classified them as Keep, Revise, or Investigate, and checked the Space, paper, and poster for consistency.

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A Technical Details and Reproducibility

Model checks and the completed synthetic experiment are recorded below; human behavioral evidence remains planned. The final A0 poster is `FP08_Final_A0_Poster.pdf`.

A.1 Model definitions and analytical checks

The synthetic scoring rule is $r_i = 100E_iS_i$, with normalized exposure E_i and downside severity S_i in $[0, 1]$. This is a policy score, not estimated research benefit. Freeze portfolios and requests; exclude duplicate requests, resale, and within-batch score manipulation. Credits have unit marginal opportunity cost. Correlated financial outcomes and endogenous portfolio choice are outside the baseline.

For the independent, uniform, single-slot benchmark, a type-v bidder mimicking type z against $\beta(z) = (n-1)z/n$ has expected utility $U(z; v) = [v - (n-1)z/n]z^{n-1}$. Its derivative is $(n-1)z^{n-2}(v-z)$, maximized at $z = v$. Bids above the maximum equilibrium bid only raise payment; abstention gives zero. This verifies only the separate single-slot first-price benchmark used for theoretical and behavioral comparison; it is not claimed to be the exact equilibrium of the multi-slot hybrid mechanism. In the independent baseline, public risk scores do not update beliefs about private values because v_i and r_i are generated independently. Correlated value–risk environments are examined only in sensitivity analysis.

For any realized allocations, $W(H) - W(P) = -L + \lambda\Delta C$. If $L, \Delta C > 0$, improvement requires $\lambda > L/\Delta C$; if $L \geq 0$ and $\Delta C \leq 0$, strict improvement is impossible. As an arithmetic illustration, replacing $(v, r) = (90, 20)$ with $(60, 90)$ gives $L = 30$, $\Delta C = 70$, and threshold $3/7$. At $\lambda = 0.5$ the gain is 5; at $\lambda = 0.2$ it is -16. This is not a simulation result. A pure first-price auction maximizes private value only when bids rank values correctly.

A.2 Computational inputs, algorithm, and verification

The fixed synthetic design uses $N = 20$, $K = 4$, $M = 100$, $R = 1$, $r^* = 60$, and $\lambda = 1$. Independently draw $v \sim U[0, 100]$ and exposure/severity $E, S \sim U[0, 1]$; freeze $r = 100ES$ before bidding. Use an independent random arrival permutation. The assumed bid $b = 0.8v$ is identical across mechanisms, including after the reserve outcome; it does not solve the multi-slot equilibrium. Budgets are nonbinding under these bids. Because every PM begins with the same 100-credit endowment, including unused credits gives total payoff $100 + x_i v_i - p_i$. The constant 100 does not affect strategic comparisons, so we report normalized utility

$$u_i = x_i v_i - p_i.$$

. Repeated draws are independent Monte Carlo replications, not strategic rounds.

Language-independent pseudocode. For each batch, freeze values, scores, arrival, and assumed bids. Pure first-price auction selects the top four bids and charges each its own bid. Hybrid selects up to R highest-risk eligible requests at zero payment, removes those PMs, then fills remaining capacity by first-price bids; unused reserves return to the auction. FCFS selects four by arrival and charges zero. Lower PM ID breaks all exact ties. Stop after allocation and payment; record each winner once. Compute paired differences on the same requests, then average across batches.

Metrics. Record winner IDs and total payments, V, C, W, and total utility $V - \sum p_i$. For the predetermined first batch, also report all individual payments and utilities. Value efficiency divides V by the top-four value sum. Eligible access divides eligible winners by eligible requests; no-eligible batches are NA and excluded from its mean. These metrics are distinct: risk coverage does not estimate avoided losses.

Fresh run and final verification. The recorded computational run was generated on September 26, 2026, using Python 3.9.6, with 1,000 primary batches and seed 20260926. A final clean verification run was completed on October 20, 2026. No third-party packages are required. Running `simulation.py` followed by `verify_reproduction.py` executes six tests and reproduces 31,265 CSV cells in a clean temporary directory with numerical tolerance 10^{-10}.

Sensitivity and limits. The 36 scenarios vary the risk threshold over $r^* \in \{40, 60, 80\}$ and the reserve size over $R \in \{1, 2\}$. They also vary three value–risk alignments and two bid rules. Alignment stress tests sort the same values with or against risk; heterogeneous shading uses independent $b_i = \alpha_i v_i$, with $\alpha_i \sim U[0.5, 1]$, fixed across mechanisms. Welfare weights are 0, 0.5, 1, 2. At the primary threshold/reserve, common shading gives mean $\Delta W = 0$ under positive rank alignment and −3.24 under negative rank alignment; heterogeneous shading gives +13.12 in the independent case. These are hypothetical behaviors, not peer observations. Under the primary design, only 43.8% of batches strictly improve W at $\lambda = 1$; the positive mean is not dominance. The paired Monte Carlo SE is 0.89, measuring simulation sampling uncertainty only. The break-even weight 0.733 is a ratio of mean loss to mean coverage gain, not a mean of batch thresholds. The design excludes strategic score manipulation, common values, endogenous arrival, and future decisions. Human-bid import and Hugging Face parity tests remain planned.

A.3 AI Use Disclosure

OpenAI ChatGPT and Codex were used to support model clarification, literature checking, LaTeX editing, simulation implementation, debugging, and artifact drafting.

ChatGPT materially assisted with clarifying the distinction between the single-slot Bayesian benchmark and the stylized multi-slot bidding rule, refining the reserve-allocation sequence, separating internal payments from firm-wide welfare, and narrowing the novelty claim. The authors reviewed these suggestions and retained only those consistent with the course requirements and implemented model.

Codex assisted with reviewing draft sections, checking model and literature consistency, implementing and debugging the synthetic simulation, generating reproducibility checks, and reproducing archived outputs. Six automated tests and a clean rerun were used to verify the reported computational results.

No AI-generated output was accepted as evidence without author review. The authors independently selected the final research question, mechanism, parameterization, welfare definition, and behavioral design, and remain responsible for all claims, citations, computations, and limitations.

B Cumulative Intellectual Development

The project began as a general internal-auction problem in which portfolio managers competed for scarce AI research capacity using private bids. This initial framing revealed demand but did not clearly distinguish the project from existing resource-allocation applications.

During team development in September 2026, we revised the model by introducing a separate risk-reserve channel based on an auditable, bid-independent downside-risk score. This changed the research question from whether auctions can allocate scarce AI resources to when private willingness to pay should be supplemented by independently verified risk protection.

A second revision separated the exact single-slot first-price Bayesian benchmark from the multi-slot computational experiment. The equilibrium formula $b(v) = (n-1)v/n$ is now used only for the single-slot theoretical and behavioral benchmark, while the multi-slot simulation explicitly labels $b_i = 0.8v_i$ as an assumed monotone bidding rule.

A third revision separated internal payments from firm-wide welfare and defined the main institutional trade-off as

$$W(H) - W(P) = -L + \lambda \Delta C.$$

The resulting PS2 contribution therefore shifted from applying an auction to a new AI setting toward testing when a hybrid auction-plus-reserve mechanism can improve the balance between private-value efficiency and verified downside-risk coverage.

Table 1: Development roadmap.

Horizon	Milestone
Foundation	Bayesian games and auction design.
PS2	Conditional welfare test, 1,000 matched batches, and linked artifacts.
Symposium	Present the bounded claim and record peer challenges.
Next study	Framing experiment, human-bid import, and score audit.
2056	Accountable allocation of scarce AI capacity under private demand and verifiable risk.

Retained PS2 development roadmap

C Open Source and Three Lens Inspiration

Software and source record. The completed computational experiment uses Python 3.9.6 and its standard library (official source: https://www.python.org/; PSF License). This choice supports a lightweight and reproducible simulation without third-party packages, but it does not validate the economic assumptions or establish external validity.

The behavioral artifact is hosted as a static Hugging Face Space at https://huggingface.co/spaces/dku-comsci-econ206-2026/Final_Zhenning_Qianshuo_PM_Bid. It uses browser-side HTML, CSS, and JavaScript to structure the decision setting, benchmark comparison, reflection, and peer play. The Space is an interaction interface rather than a source of population-level evidence; the archived computational outputs remain the source of the reported 1,000-batch mechanism results.

The computational repository is maintained at GitHub, with the recorded PS2 commit `59b1ead`. The runnable notebook and reproduction files document the implemented model, parameters, outputs, and limitations.

The runnable Colab notebook is available at Colab.

The interdisciplinary artifact is an A0 landscape poster that integrates the shared research question, verified gap, strategic model, three lenses, computational comparison, behavioral evidence status, real-world case, limitations, and links to the GitHub and Hugging Face artifacts. The editable poster file is `FP08_Final_A0_Poster.pptx`, and the final PDF is `FP08_Final_A0_Poster.pdf`.

The poster uses the same research question, baseline parameters, numerical results, mechanism definitions, limitations, instructor information, acknowledgements, and artifact links as the paper. All analytical charts, tables, and mechanism diagrams are team-generated from the reported synthetic results. The Duke Kunshan University logo is an external institutional visual.

Intellectual inspirations. Harsanyi's Bayesian-games framework motivates the incomplete-information solution concept used for the single-slot bidding benchmark, while **Vickrey**'s auction analysis motivates strategic bid shading. Work on internal compute markets and quota-market allocation motivates treating high-capability AI access as a scarce allocable resource, while reserve systems motivate protected non-price access. Related GenAI asset-management work establishes the application context. These sources motivate components of the design but do not directly answer when a verified current-risk reserve should displace private-value allocation in portfolio research.

Three-lens consequence. *Game theory* changes the prediction: private values and simultaneous bidding generate strategic bid shading rather than truthful reporting, with the exact equilibrium benchmark restricted to the separate single-slot case.

Social choice changes the evaluation: allocation quality is judged not only by private research value V but also by independently verified downside-risk coverage C, making the welfare weight λ explicit.

Mechanism design changes access: FCFS ignores both value and risk, the pure first-price auction allocates all capacity through bids, whereas the hybrid first-price auction with a risk reserve first protects one high-risk eligible request and then allocates the remaining capacity through first-price bidding.

Together, the three lenses change the prediction, evaluation criterion, and allocation rule rather than applying three labels to an unchanged mechanism.

D Feedback Response and Revision Record

Review-ready version. The September 27 review-ready version uses the portfolio-manager AI research allocation model with $N = 20$, $K = 4$, one verified risk-reserve slot, and three comparison rules: FCFS, a pure first-price auction, and a hybrid first-price auction with one risk reserve. The computational baseline, sensitivity checks, artifact links, and current evidence status are frozen for review.

Pre-symposium revisions. Before the review-ready submission, the team revised the project in three material ways. First, we narrowed the contribution from a generic AI-quota auction to a portfolio-research setting in which private willingness to pay may not fully reflect verified downside risk. Second, we separated the exact single-slot first-price Bayesian benchmark from the assumed multi-slot bidding rule used in simulation. Third, we separated internal payments from firm-wide welfare and made the main trade-off explicit as private-value efficiency versus verified downside-risk coverage.

D.1 Formal response to September 28 feedback

D.1.1 Poster readability and visual density. **Source and point raised. Temur Akhtamjonov** noted that the Hugging Face game design was a strength, but that the A0 poster was visually crowded and some elements appeared too compressed. **Gihun Lee** similarly found the poster and repository well organized but indicated that some figures and tables would benefit from clearer presentation and captions.

Decision and reason: Revise. We accepted this feedback because the concern affects whether the strategic model, mechanism comparison, and computational evidence can be understood quickly during a poster presentation, even though it does not change the underlying model or results.

Change made. We reduced text density in the poster, increased spacing between major sections, shortened explanatory text around figures, and revised figure and table captions so that the displayed comparison can be interpreted without relying on the oral presentation. We retained the same model assumptions, mechanisms, parameters, and numerical results.

Evidence location. See the revised PS2 A0 poster, especially the mechanism-comparison and computational-results panels. The underlying computational evidence remains unchanged and is reproduced in Section 3 and Appendix A.2 of the PS2 PDF, and in the GitHub repository at commit `59b1ead`.

Remaining question. Poster readability has been revised, but further visual simplification could be tested with readers unfamiliar with the project. We treat this as a communication improvement rather than a change to the research design.

D.1.2 Interpretation of the break-even welfare weight. **Source and point raised. Gihun Lee** asked how the reported break-even value of approximately $\lambda = 0.733$ should be interpreted in a real asset-management setting.

Decision and reason: Revise. We accepted this concern because the original presentation could be misread as implying that 0.733 is an empirically estimated firm preference. It is instead a model-implied threshold under our synthetic calibration.

Change made. We clarified that the break-even welfare weight is calculated from the simulated mean private-value loss and mean risk-coverage gain, $\lambda^* = L/\Delta C$, and therefore indicates how much weight the stated welfare function must place on one unit of verified risk coverage, relative to one unit of private research value, for the hybrid mechanism to break even against the pure auction. The value 0.733 is not an estimate of any real firm's preferences and should not be interpreted as a directly observed policy parameter.

Evidence location. The interpretation is stated in PS2 Section 3 and Appendix A.2. The primary simulation gives a mean private-value loss of 26.82, a mean risk-coverage gain of 36.59, and a corresponding break-even ratio of approximately 0.733.

Remaining question. A real deployment would require the firm to justify or estimate an operational welfare weight using institutional objectives or observed decision outcomes. Estimating such a parameter from real organizational data remains planned future work.

D.1.3 Assumed bidding rule and endogenous strategic response. **Source and point raised. Longfei Jing** observed that the trade-off between private value and risk coverage was clearly quantified, but noted that the computational bidding rule $b_i = 0.8v_i$ is assumed rather than derived from equilibrium. She asked whether the hybrid mechanism would still outperform the pure auction if bidders changed their strategies in response to the reserve rule.

Decision and reason: Investigate. We agree that endogenous strategic response is an important limitation. However, solving the exact equilibrium of the multi-slot hybrid mechanism would substantially expand the scope of the current PS2. We therefore do not reinterpret the assumed multi-slot bidding rule as an equilibrium result. Instead, we narrow the claim and test whether the welfare comparison is robust to alternative stylized bidding behavior.

Change made. We explicitly separate the exact single-slot first-price Bayesian Nash equilibrium benchmark, $b(v) = (n-1)v/n$, from the multi-slot simulation, where $b_i = 0.8v_i$ is only an assumed monotone bidding rule. We also report a heterogeneous-shading sensitivity condition, $b_i = \alpha_i v_i$, $\alpha_i \sim U[0.5, 1]$, held fixed across mechanisms.

Evidence location. The distinction between the equilibrium benchmark and the assumed simulation rule appears in PS2 Section 2, Section 3, Appendix A.1, and Appendix A.2. The heterogeneous-shading sensitivity results are reported in Appendix A.2 and reproduced in the GitHub simulation outputs at commit `59b1ead`.

Remaining question. The current project does not solve the endogenous multi-slot equilibrium of the hybrid mechanism. Whether reserve eligibility changes equilibrium bidding strategies, and whether such strategic adaptation changes the welfare ranking, remains planned future work.

D.1.4 Behavioral validity of the assumed bid rule. **Source and point raised. Jiayang Sun** similarly noted that the welfare comparison depends on the assumed bidding rule and asked whether real participants would bid according to the modeled relationship between value and bid. She suggested running the Hugging Face behavioral interaction with human participants and comparing observed bids with the theoretical benchmark.

Decision and reason: Revise and Investigate. We accepted the interpretation concern and retained the behavioral test as an unresolved empirical question. Because no completed human-bidding round was available at the time of the final computational comparison, we do not present simulated bids as observed behavioral evidence.

Change made. We revised the paper and behavioral artifact to label $b_i = 0.8v_i$ as a computational assumption rather than an observed behavioral rule. The Hugging Face Space retains a separate four-bidder, single-slot task in which participants can submit bids under matched gain and loss frames and compare their choices with the rational benchmark $b(v) = 3v/4$.

Evidence location. See PS2 Section 4 for the planned behavioral comparison and Appendix F.1 for the behavioral protocol. The Hugging Face Space is https://huggingface.co/spaces/dku-comsci-econ206-2026/Final_Zhenning_Qianshuo_PM_Bid.

Remaining question. Human bidding evidence has not yet established whether participants systematically follow, overbid, or underbid the rational benchmark, or whether gain/loss framing changes bids while numerical payoffs remain fixed. This remains planned exploratory work and is not used to support the completed welfare result.

E Structured Abstract and Cumulative Proposal Map

Table 2: Final eight-part proposal map and revision evidence.

Part	Current statement and revision evidence
Background	Internal compute markets and reserve systems already exist; the project examines verified risk reserves alongside paid bids for scarce AI research.
Motivation	Private research demand may omit institution-wide risk priorities; reserve adoption requires evidence about displaced value and added coverage.
Application Scenario	Twenty PMs compete for four slots at a modeled asset manager during earnings season; these constraints are research assumptions.
Research Question	When does risk protection justify displaced value? Compare three mechanisms and probe assumed bids through planned framing research.
Method	Public frozen risk, private values, unit demand, FCFS/pure/hybrid allocation, assumed $b_i = 0.8v_i$, and paired synthetic batches.
Results	Across 1,000 matched synthetic batches, mean $\Delta W = 9.78$ at $\lambda = 1$ and -8.52 at $\lambda = 0.5$, with a break-even weight of $\lambda^* \approx 0.733$. Final robustness checks include a fixed-batch bid-shading perturbation and a risk-score threshold perturbation around $r^* = 60$.
Intellectual Merits	The final three-paper comparison positions FP08 relative to Athey et al., Sivan et al., and Karma. The bounded contribution makes the trade-off between displaced private value and verified risk coverage explicit; Figure 1 summarizes the inherited foundations, contribution, supporting evidence, and limitations.
Practical Impacts	The final impact pathway reports simulated allocation indicators, intended links to SDG Targets 9.5 and 16.6, possible harms, and required safeguards. Figure 2 uses official SDG icons and distinguishes demonstrated simulation results from intended institutional and financial impacts.

Retained PS2 structured abstract

F Retained PS2 Auction Artifact and Symposium Record

F.1 Auction laboratory record

The completed computational auction uses the inputs, allocation rules, payments, and outputs reported in Appendix A.2. The separate human behavioral round has not yet been conducted. Its planned identifier is `HF-FP8-single-slot-v1`. Four participants compete for one slot under an assigned gain or loss frame. Each participant receives a private value in $[0, 100]$ and submits one sealed bid in $[0, 100]$. The highest bidder wins, pays their own bid, and the game stops. The rational benchmark is $b(v) = 3v/4$.

The Hugging Face template records the assigned frame, private value, actual bid, benchmark deviation, winner, payment, utility, revenue, allocation efficiency, confidence, reflection, and resulting Keep/Qualify/Revise decision.

F.2 Linked-artifact parity

Table 4: Linked-artifact parity record.

Artifact	Parity check
GitHub	Commit `59b1ead`; model, executed notebook, outputs, and clean rerun are recorded in Appendix A.2. https://github.com/dku-comsci-econ206-Autumn2026/PS2_FP8_Qianshuo_Zhenning
Hugging Face	Live static Space at https://huggingface.co/spaces/dku-comsci-econ206-2026/Final_Zhenning_Qianshuo_PM_Bid. It uses the same decision setting, three mechanisms, behavioral benchmark, planned evidence status, and privacy boundary as the proposal.
A0 poster	Editable poster file: `FP08_Final_A0_Poster.pptx`. Final PDF: `FP08_Final_A0_Poster.pdf`. The poster uses the same research question, mechanism definitions, baseline numerical results, real-world case, limitations, instructor information, acknowledgements, artifact links, and selected references as the paper.

F.3 Symposium and practical-impact record

The review-ready claim for the symposium is that, under the stated primary parameters, the hybrid mechanism increases mean risk coverage enough to produce a positive mean ΔW at $\lambda = 1$. This is not universal dominance: the result depends on the

Proposal element	Current statement / change
Background	Internal compute markets and reserve systems already exist. This project examines whether a verified risk reserve can improve the allocation of scarce AI research capacity at an asset manager relative to a pure first-price auction.
Motivation and application	During earnings season, portfolio managers may bid according to private research value even when some requests carry broader verified downside risk for exposed portfolios. Competing PMs, the firm risk team, management, and clients are affected.
Three connected questions	Q1 asks when protecting verified risk justifies displaced private value. Q2 compares FCFS, a pure first-price auction, and a hybrid first-price auction with a risk reserve on matched computational batches. Q3 tests whether human bidding departs from the rational single-slot benchmark under fixed-payoff gain/loss framing. The shared question requires all three answers.
Method and model assumptions	Twenty unit-demand PMs compete for four premium AI research slots. Each PM has a private research value, while the firm observes a public, frozen, bid-independent risk score. In the hybrid mechanism, one eligible high-risk request receives the reserve slot before the remaining capacity is allocated through first-price bidding. The exact Bayesian Nash equilibrium benchmark is used only for the separate single-slot case; the multi-slot simulation uses an assumed monotone bid rule $b_i = 0.8v_i$.
Results or expected results	Analytically, the hybrid improves the stated welfare index when $\lambda\Delta C > L$. In the 1,000-batch synthetic comparison, the hybrid gains 36.59 units of risk coverage while losing 26.82 units of private value relative to the pure first-price auction. Mean $\Delta W = 9.78$ at $\lambda = 1$ and -8.52 at $\lambda = 0.5$, with a break-even welfare weight of approximately 0.733, computed as the ratio of mean private-value loss to mean risk-coverage gain.
Intellectual merits	The project does not claim novelty for auctions, reserve systems, or AI resource markets separately. Its contribution is a bounded application-level comparison that combines private strategic bids with an independent, verifiable risk-protection channel and makes the cost of displaced private value explicit.
Practical impacts	The proposed mechanism may help asset managers make explicit the trade-off between private research demand and verified downside-risk protection. Potential beneficiaries include portfolio managers, risk teams, management, and clients. Risks include mismeasured risk scores, systematic access bias, strategic gaming outside the modeled batch, and overreliance on synthetic calibration. Real-world use would require audited risk inputs, transparent eligibility rules, access monitoring, and evidence linking AI research priority to actual decision quality or downside outcomes.
Feedback received and revision made	Before symposium review, the team revised the project from a generic AI-quota auction to a portfolio-research mechanism with a verified risk reserve, separated the exact single-slot Bayesian benchmark from the assumed multi-slot bidding rule, and treated internal payments as transfers rather than firm-wide welfare losses. September 28 peer feedback identified three main revision priorities: poster readability, interpretation of the break-even welfare threshold, and the limitation of the assumed multi-slot bidding rule. The final PS2 revision clarifies these points, preserves the existing verified simulation, and records endogenous and human bidding as planned extensions.

Table 3: Appendix E structured abstract and cumulative proposal map.

welfare weight, value–risk alignment, risk-score validity, and assumed bids. The real-world case therefore requires score audits, decision records, access monitoring, and a correction route.

Symposium feedback and final PS2 revision. The September 28 peer reviews raised three main issues: poster readability, interpretation of the break-even welfare weight, and the reliance of the multi-slot simulation on an assumed bidding rule. **Temur Akhtamjonov** and **Gihun Lee** motivated revisions to poster presentation and explanation of the welfare threshold. **Longfei Jing** and **Jiayang Sun** questioned whether the assumed rule $b_i = 0.8v_i$ would remain appropriate under endogenous or human bidding.

The team classified the presentation and interpretation issues as Revise, and the endogenous-bidding question as Investigate. We clarified that $\lambda = 0.733$ is a synthetic break-even threshold rather than a real-world estimated preference parameter, strengthened the distinction between the exact single-slot BNE benchmark and the assumed multi-slot bidding rule, and retained heterogeneous bid shading as a robustness check. Human bidding under the Hugging Face protocol remains planned exploratory work.

No symposium comment changed the implemented baseline parameters or primary mechanism definitions. Therefore, the primary simulation was not replaced; the existing clean rerun and sensitivity analysis remain the verified computational evidence. The final PS2 PDF, poster, GitHub repository, and Hugging Face Space were checked for consistent mechanism definitions, evidence labels, numerical claims, and limitations.

Retained planned behavioral comparison

Q3 is a planned human-bidding task, distinct from the completed Colab simulation. Four bidders compete for one slot in a first-price auction. With symmetric risk-neutral bidders and independent uniform values, the benchmark is $b(v) = 3v/4$. Gain and loss frames use identical numerical payoffs and differ only in wording. Our Space (https://huggingface.co/spaces/dku-comsci-econ206-2026/Final_Zhenning_Qianshuo_PM_Bid) structures the decision, self-reflection, and four-person peer play without storing responses. Higher loss-frame bids would qualify the benchmark; similar bids near $0.75v$ would retain it; departures in both frames would motivate revision of the bid rule or interface. Risk attitude, misunderstanding, and beliefs about opponents remain competing explanations. No human round has been completed, so behavioral evidence remains planned and exploratory.

G Final Week Evidence and Final Release Audit

G.1 October 12 Beginning Post and Duet Investigation

Beginning Post Oct 12: *FP8, Auctions with Risk Reserves for Scarce AI Research: Beginning Post Oct 12.*

Beginning-post URL: *https://edstem.org/us/courses/102917/discussion/8381683?comment=19460314.*

Post version or timestamp: *Title: FP8, Auctions with Risk Reserves for Scarce AI Research -Beginning Post Oct 12*

Nobel Frontier: Our expected Nobel research-frontier contribution is to develop a welfare-based mechanism for allocating scarce AI research capacity, balancing private research value with verifiable downside risk. This frontier matters because efficient market-based allocation alone may overlook broader institutional risks, and our research seeks to identify when risk protection justifies the value displaced.

Team members: Qianshuo (Aaron) Wang and Zhenning Wang.

GitHub repository: https://github.com/dku-comsci-econ206-Autumn2026/PS2_FP8_Qianshuo_Zhenning.

Tested computational commit at that stage: `59b1ead`.

Current Hugging Face Space: https://huggingface.co/spaces/dku-comsci-econ206-2026/Final_Zhenning_Qianshuo_PM_Bid.

October 12 poster copy: https://static.us.edusercontent.com/files/lj2DsjH6VH4RnSN247kjFIxY.

Final GitHub-hosted poster URL, if later updated: `FP08_Final_A0_Poster.pdf`; repository-hosted download URL pending publication.

Beginning-post claim. The post stated a bounded research-frontier claim: a welfare-based mechanism for allocating scarce AI research capacity should make explicit the tradeoff between private research value and verifiable institutional risk. The project does not claim that auctions, reserves, or internal AI-resource markets are individually new.

Duet partner. Our duet partner was FP5: **Temur Akhtamjonov**, **Mustafa Ayub Khan**, and **Gihun Lee**.

Duet investigation or discussion URL: *https://edstem.org/us/courses/102917/discussion/8381683?comment=19460476.*

Three closest studies. The joint literature investigation compared FP08 with:

(1) Athey, Coey, and Levin's analysis of auction set-asides and subsidies [1];

(2) Sivan et al.'s dynamic quota market for scarce machine-learning resources [5];

(3) Vuppalapati et al.'s Karma mechanism for repeated, dynamic resource demand [13].

Our response distinguished FP08 from each study. Relative to Athey et al., eligibility is based on verified current risk rather than bidder identity, although a subsidy remains an untested counterfactual. Relative to Sivan et al., FP08 adds a separate risk-coverage objective and reserves one of four slots instead of allocating all capacity through dynamic prices. Relative to Karma, the present model is one-shot and does not include allocation history, carryover credits, or repeated strategic demand.

Recorded defense-response URL: *https://edstem.org/us/courses/102917/discussion/8381683?comment=19460439.*

Division of work. Qianshuo Wang led the strategic model, welfare condition, computational mechanism comparison, and reproducibility record. Zhenning Wang led the behavioral evidence boundary, practical-impact analysis, Hugging Face interface, acknowledgments, and future roadmap. Both authors contributed to the shared question, interpretation, poster, and final integration.

October 12 takeaway. The reserve is not universally superior. Under the stated welfare function, it is justified only when

$$\lambda \Delta C > L.$$

The result remains conditional on strategic bidding, risk-score validity, and the institutional interpretation of risk coverage.

October 12 takeaway or reflection URL: *https://edstem.org/us/courses/102917/discussion/8381683?comment=19460571.*

G.2 October 14 Feedback, Self-Challenge, and Integration

October 14 final-post or discussion URL: *https://edstem.org/us/courses/102917/discussion/8391519?comment=19481089.*

Final poster evidence location: `FP08_Final_A0_Poster.pdf`.

Poster revision. Temur Akhtamjonov and Gihun Lee identified poster density, figure presentation, and interpretation of the welfare threshold as revision priorities. We reduced text density, strengthened captions, and clarified that the break-even value $\lambda^* \approx 0.733$ is a synthetic model threshold rather than an estimate of a real firm's preferences.

Strategic-bidding challenge. Longfei Jing and Jiayang Sun challenged the assumed multi-slot rule $b_i = 0.8v_i$. We separated the exact single-slot Bayesian benchmark

$$b(v) = \frac{n-1}{n}v$$

from the assumed multi-slot rule. We retained heterogeneous shading as a sensitivity check and identified endogenous strategic bidding and observed human bidding as future tests.

Duet implementation question. FP5 asked:

> If you were to apply your model at DKU, what would be the most difficult part?

Our response was that the principal difficulty would be defining a priority or risk measure that is observable, auditable, and accepted as legitimate. The auction can be implemented mechanically, but different users may have different reasons for requesting scarce AI capacity, and those needs may not be summarized accurately by one score.

Three research challenges. The final self-challenge contains three connected limitations:

(1) **Economics:** the multi-slot comparison uses assumed bidding rather than an endogenous equilibrium;
(2) **Computer science and governance:** the risk score and eligibility threshold have not been validated using institutional data or a correction procedure;
(3) **Behavioral science:** the gain/loss framing experiment remains planned, so simulated bids are not evidence of human behavior.

These challenges led to three proposed studies: an endogenous-bidding benchmark, a risk-score audit with threshold perturbation and correction, and preregistered matched human bidding rounds.

Individual self-challenge URL: *https://edstem.org/us/courses/102917/discussion/8391519?comment=19481123.*
Assigned peer-reply: *Replied in in-class discussion.*

G.3 Integration of the Three Final Diagrams

The three required diagrams are integrated into the main paper rather than repeated in this appendix:

- Figure 1, presented at the beginning of the paper, summarizes the three closest studies, FP08's distinct contribution, supporting evidence, and main limitations;
- Figure 2 in Section 8 connects the mechanism to stakeholders, simulated indicators, SDG Targets 9.5 and 16.6, possible harms, and safeguards;
- Figure 3 in Section 8 connects each unresolved assumption to a question, method, possible collaborator, progress criterion, and milestone.

The intellectual-merit diagram distinguishes inherited foundations from the project's bounded contribution. The practical-impact diagram separates demonstrated simulated allocation effects from intended SDG and financial benefits. The future roadmap treats endogenous bidding, score validity, and human behavior as required evidence before institutional adoption.

Full references for the closest literature and behavioral framing foundation appear in the bibliography [1, 5, 9, 13]. SDG Targets 9.5 and 16.6 are supported by the official United Nations goal pages [10, 11].

Section 7 diagram verification: *Zhenning Wang.*
Section 8 impact-diagram verification: *Zhenning Wang.*
Section 8 roadmap-diagram verification: *Qianshuo(Aaron) Wang.*
Official SDG icon status: *Inserted.*

G.3.1 Expanded Future Research Roadmap. Table 5 expands the compact roadmap in Figure 3. Each study converts a current limitation into a specific research question, method and data requirement, possible collaborator, progress criterion, and milestone. These studies are planned extensions rather than completed evidence.

Table 5: Expanded future research roadmap from current limitations to evidence required before institutional adoption.

Lens	Research Question	Method / Data	Possible Collaborator	Progress Criterion	Milestone
Economics	Does the Hybrid-versus-Pure-FPA welfare ranking survive when multi-slot bidding is endogenous rather than fixed at $b_i = 0.8v_i$?	Develop a strategic multi-slot bidding benchmark and compare equilibrium or best-response behavior with the current matched synthetic batches and stylized-bid sensitivity results.	Auction / mechanism-design researcher	Characterize strategic bidding under the reserve rule and determine whether the sign and break-even condition for ΔW remain robust.	Replace the fixed-bid assumption with a validated strategic benchmark and report the revised welfare comparison.
Computer Science	Is the risk score sufficiently valid and stable for reserve eligibility, especially near the threshold $r^* = 60$?	Validate and calibrate the score using frozen institutional inputs; audit measurement error and classification around $r^* = 60$; test correction and appeal procedures.	Risk-model validation, data-science, or compliance team	Show acceptable score reliability and allocation stability under plausible measurement perturbations, with documented correction procedures for borderline cases.	Complete a score-validation and threshold-audit protocol before any shadow deployment.
Behavioral Science	Does gain/loss framing change human bid shading when numerical payoffs are held fixed?	Run preregistered matched human bidding rounds using the existing four-bidder, single-slot Hugging Face task and compare bids with the benchmark $b(v) = 3v/4$.	Behavioral-economics researcher or experimental-lab partner	Collect sufficient valid observations to estimate whether bid deviations differ systematically across frames and whether behavior departs from the benchmark.	Complete the first human-bidding study and use the results to retain, qualify, or revise the behavioral assumptions.

G.4 Poster and Repository Release Record

The root GitHub README displays the GitHub-hosted final poster preview and provides a link to the committed PDF.
Final poster PNG repository path: `poster/FP08_Final_A0_Poster.png`.
Final poster PDF filename: `FP08_Final_A0_Poster.pdf`.

Final poster PNG URL:
https://github.com/dku-comsci-econ206-Autumn2026/PS2_FP8_Qianshuo_Zhenning/blob/main/poster/FP08_Final_A0_Poster.png

Final poster PDF URL:
https://github.com/dku-comsci-econ206-Autumn2026/PS2_FP8_Qianshuo_Zhenning/blob/main/poster/FP08_Final_A0_Poster.pdf

Root README URL:
https://github.com/dku-comsci-econ206-Autumn2026/PS2_FP8_Qianshuo_Zhenning/blob/main/README.md

README poster-embed verification: Checked by Qianshuo (Aaron) Wang on October 20, 2026.
Poster source or editable version: `FP08_Final_A0_Poster.pptx`.
Historical PS2 Canva link (not the final editable source):
https://canva.link/vlx5ljli23z7qk8

Official United Nations SDG communication materials:
https://www.un.org/sustainabledevelopment/news/communications-material/

Field-learning reference:
https://cscc.dukekunshan.edu.cn/where-scholarship-industry-and-public-science-meetfield-based-learning-and-interdisciplinary-dialogue-for-future-global-leaders/

G.5 Final Version Audit

Audit item	Recorded status
GitHub repository	https://github.com/dku-comsci-econ206-Autumn2026/PS2_FP8_Qianshuo_Zhenning.
Tested PS2 commit	`59b1ead`. This is the recorded computational commit used for the reproduced PS2 results.
Final GitHub commit or release	`ac2fbf5`. Final synchronized repository commit: https://github.com/dku-comsci-econ206-Autumn2026/PS2_FP8_Qianshuo_Zhenning/commit/ac2fbf5eb3da11543c49f481aea257eaee097373.
Final fresh-run date	October 20, 2026.
Final fresh-run result	Six computational tests passed. The reproduced PS2 clean run verified 31,265 CSV cells with tolerance 10^{-10} and reproduced the recorded computational outputs.
Runnable notebook	https://colab.research.google.com/drive/1I-J52wxGaAIwJK_LjtghBnXS2GxY0U2V.
Final notebook run evidence	Final computational run completed on October 20, 2026. The runnable Colab notebook above contains the final computational workflow; a separate archived run record was not created.
Verified computational outputs	The final clean run contains 1,000 primary matched batches. Six tests reproduced 31,265 CSV cells with tolerance 10^{-10}, consistent with the recorded PS2 computational results.
Final Hugging Face Space	https://huggingface.co/spaces/dku-comsci-econ206-2026/Final_Zhenning_Qianshuo_PM_Bid.
Hugging Face final version	*f6b1ef4*.
Poster preview and PDF	PNG: https://github.com/dku-comsci-econ206-Autumn2026/PS2_FP8_Qianshuo_Zhenning/blob/main/poster/FP08_Final_A0_Poster.png PDF: https://github.com/dku-comsci-econ206-Autumn2026/PS2_FP8_Qianshuo_Zhenning/blob/main/poster/FP08_Final_A0_Poster.pdf
Paper and poster parity	*Qianshuo Wang.*
Full acknowledgments	*Qianshuo Wang & Zhenning Wang.*
Team contribution statements	*Qianshuo Wang & Zhenning Wang.*
AI disclosure	Appendix A records permitted use of ChatGPT and Codex. *OpenAI ChatGPT and Codex assisted with model clarification, literature organization, LaTeX editing, code review, and figure drafting. The authors independently verified all citations, assumptions, computations, and interpretations and remain responsible for the final work.*
Canvas files	*Complete paper PDF, complete Overleaf source ZIP, editable A0 poster, print-ready A0 poster.*
All-member confirmation	*Qianshuo Wang & Zhenning Wang.*

Table 6: Final-week evidence and release-audit record.